

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:100

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I T Sanhith (192311228) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q1. Caesar Cipher (Shift = 4)

Plaintext: MEET AT NOON

Encryption rule: Shift every alphabet forward by 4 positions.

M→Q

E→I

E→I

T→X

A→E

T→X

N→R

O→S

O→S

N→R

Ciphertext: QIIX EX RSSR

Decryption:

Shift every character backward by 4 positions.

Q→M

I→E

I→E

X→T

E→A

X→T

R→N

S→O

S→O

R→N

Recovered plaintext: MEET AT NOON

Analysis:

- Caesar cipher is a monoalphabetic substitution cipher.
- Encryption and decryption use the same key.
- Only 25 possible keys make it easy to break using brute force.

Q2. Intercepted Message

Ciphertext: GSRH RH Z HVXIVG

This matches the Atbash cipher (alphabet is reversed).

G→T

S→H

R→I

H→S

RH→IS

Z→A

HVXIVG→SECRET

Recovered plaintext:

THIS IS A SECRET

Justification:

- Atbash replaces A↔Z, B↔Y, C↔X, etc.
- Repeating letters and word structure strongly match English.
- No key is required because the mapping is fixed.

Q3. Vigenère Cipher (Keyword: KEY)

Position	Plain	Key	P Value	K Value	Cipher
1	H	K	7	10	R
2	E	E	4	4	I
3	L	Y	11	24	J
4	L	K	11	10	V
5	O	E	14	4	S

Repeated keyword: KEYKE

Ciphertext: RIJVS

Explanation:

Each plaintext letter is shifted according to the corresponding keyword letter.

Formula:

$\text{Cipher} = (\text{Plain} + \text{Key}) \bmod 26$

Q4. Digital Forensic Investigation (Steganography)

Detection Methods:

1. Statistical analysis of pixel values (histogram analysis).
2. LSB (Least Significant Bit) examination.
3. Metadata and file-size comparison.
4. Steganalysis tools such as StegExpose or zsteg.

Security Improvements:

1. Encrypt the secret message (AES) before embedding.
2. Use randomized embedding positions with a secret key instead of sequential embedding.

Analysis:

Even if hidden data is detected, encryption prevents attackers from reading it.
Random placement makes extraction significantly harder.

Q5. Rail Fence Cipher (3 Rails)

Ciphertext:

WECRLTEERDSOEEFEAOCAIVDEN

a) Decrypted plaintext:

WE ARE DISCOVERED FLEE AT ONCE

b) Vulnerabilities:

- Only changes character positions.
- Letter frequencies remain unchanged.
- Easy to attack using brute force.
- Provides very weak security by modern standards.

c) Practical Enhancement:

Combine the transposition cipher with AES encryption or encrypt the plaintext before applying transposition. This provides confidentiality even if the transposition pattern is discovered.

Tools Used

- Caesar Cipher
- Atbash Cipher
- Vigenère Cipher
- Rail Fence Cipher
- Steganography Analysis

Conclusion

This assessment demonstrates the application of classical cryptographic techniques including Caesar, Atbash, Vigenère, Rail Fence, and steganography. Classical methods help understand encryption fundamentals but are insecure for modern communication. Modern cryptography combines strong encryption algorithms such as AES and RSA with secure key management and authentication to ensure confidentiality, integrity, and authenticity.